1. Exploratory Data Analysis of Student Performance Using Education Analytics

Aim : To perform Exploratory Data Analysis (EDA) on student performance data and identify how factors such as attendance, study hours, and internal marks are associated with students' final results using statistical analysis and data visualization.

Dataset Overview

The dataset contains **20 students** with 5 attributes: Attendance (%), Study Hours (daily), Internal Marks, and Result (Pass/Fail). The data is clean — no missing values and no duplicates, so no preprocessing was required before analysis.

Key Findings

Class balance: 11 students passed (55%) and 9 failed (45%) — a reasonably balanced outcome distribution, which is good for any predictive modeling down the line.

Performance gap between Pass and Fail groups is stark and consistent across every metric:

Metric	Fail Avg	Pass Avg	Gap
Attendance	55.11%	82.45%	~27 points
Study Hours	2.11 hrs	4.36 hrs	~2.25 hrs
Internal Marks	24.33	40.00	~15.7 points

This pattern strongly suggests all three variables are meaningfully associated with the outcome — students who pass attend more, study more, and score higher internally, all at once. There's no case in the data of a student with low attendance/study hours still passing, or vice versa, which hints the relationship is fairly deterministic in this sample.

Spread of the data: Attendance has the widest variability (std dev ~15.94, ranging 45–95%), while Study Hours is comparatively tight (std dev ~1.42, ranging 1–6 hrs). Internal Marks also show wide spread (18–48), tracking closely with attendance and effort.

A soft threshold emerges: Looking at the raw rows, most students at or above ~72% attendance and 3+ study hours passed, while most below ~68% attendance and ≤3 study hours failed — though this is descriptive, not a statistically fitted cutoff.

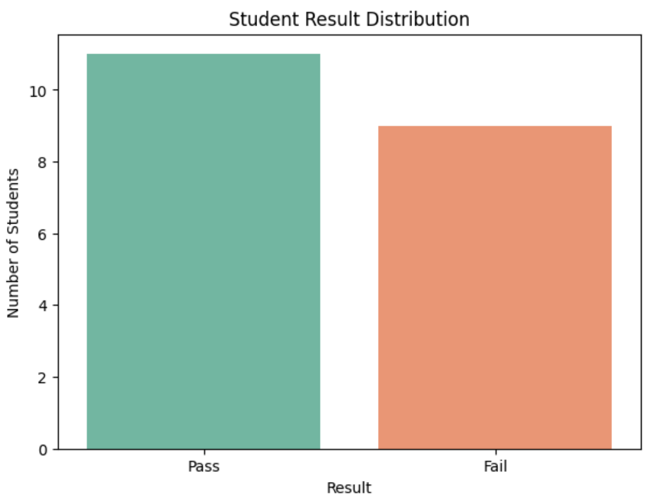
Balanced Interpretation

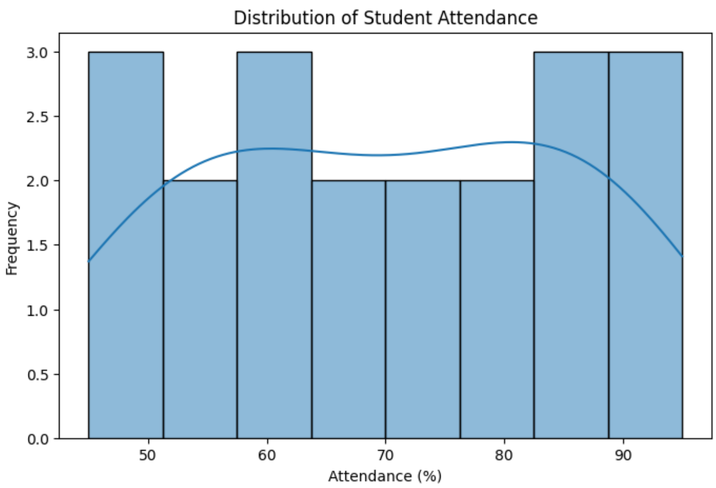
Strengths of this analysis:

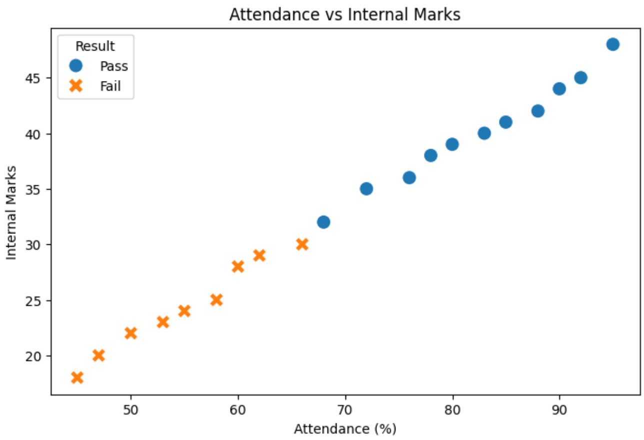
- Clean, complete data made the summary statistics reliable.
- The Pass/Fail split by group averages is a clear, intuitive way to show the relationship.

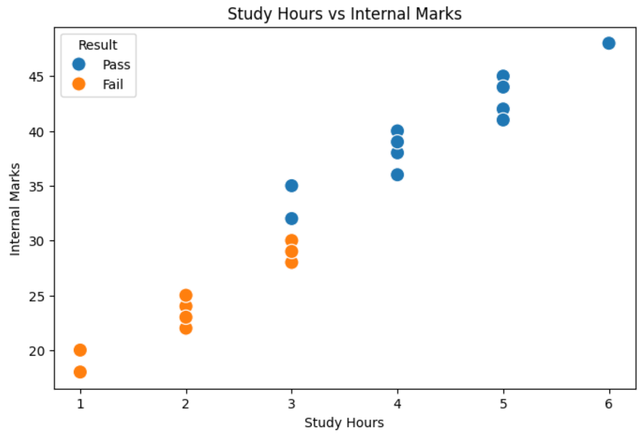
Limitations to flag:

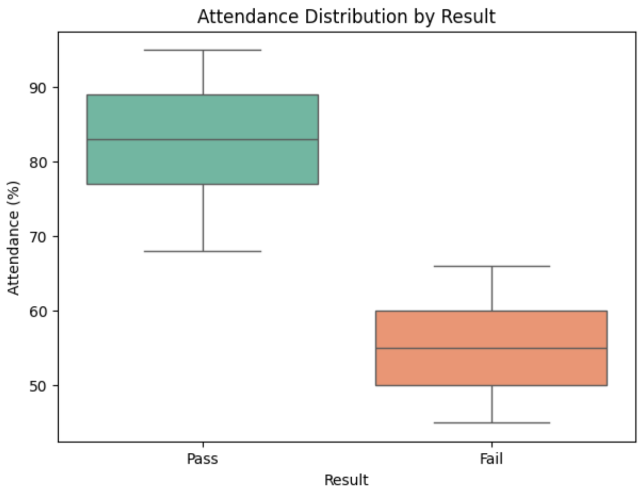
- **Sample size is small (n=20).** Averages and standard deviations from 20 rows can shift significantly with a few more data points — treat these as indicative, not conclusive.
- **Correlation ≠ causation.** Attendance, study hours, and marks all move together, but this table alone doesn't establish which factor (if any) is driving the result, or whether a hidden variable (e.g., overall diligence) explains all three.
- **No formal statistical test was run** (e.g., t-test, correlation coefficients, logistic regression) — the gap between groups is descriptive, so its statistical significance hasn't been checked.
- **Result may be a manual/rule-based label** rather than an independent outcome, given how cleanly it splits by the other three variables — worth confirming how "Result" was originally determined.

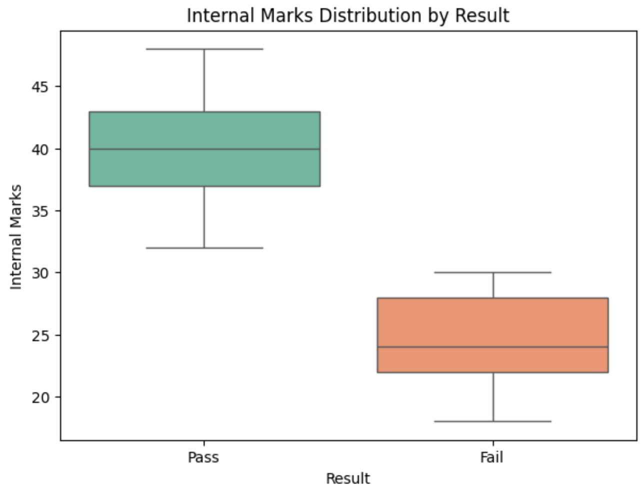


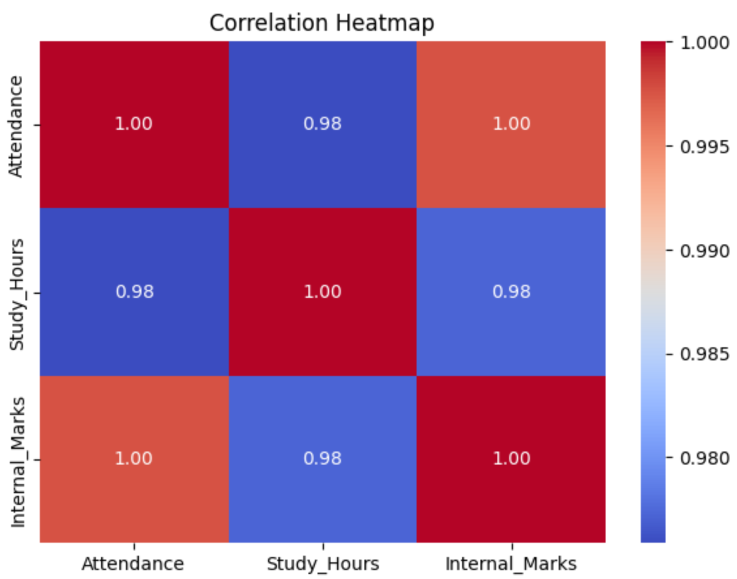












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2. Exploratory Data Analysis of Patient Health Risk Using Healthcare Analytics

Aim

To perform Exploratory Data Analysis (EDA) on patient health data and identify important patterns and relationships among age, blood sugar level, blood pressure, BMI, and disease status using statistical analysis and data visualization.

Dataset Overview

The dataset covers **20 patients** across 6 attributes: Age, Sugar Level, Blood Pressure (BP), BMI, and Disease Status (Yes/No). The data is complete and clean — no missing values, no duplicates — so it's ready for analysis as-is.

Key Findings

Class balance: A perfectly even split — 10 patients with the disease, 10 without. This is ideal for comparison and any future modeling, since there's no class imbalance to correct for.

Every health metric is markedly higher in the "Yes" (disease) group:

Metric	No Disease Avg	Disease Avg	Gap
Age	33.5	52.9	~19.4 years
Sugar Level	103.3	174.0	~70.7 points
BP	117.4	142.1	~24.7 points
BMI	23.7	30.9	~7.2 points

Sugar level shows the largest relative gap, followed by BP and BMI, with age also strongly diverging. This pattern is clinically plausible — elevated sugar, blood pressure, BMI, and older age are all well-established risk factors for conditions like diabetes and cardiovascular disease.

Variability: Sugar Level has by far the widest spread (std dev ~40.33, range 88–210), suggesting it may be the most discriminating variable between groups. BP (std dev ~14.52) and Age (std dev ~11.32) show moderate spread, while BMI is comparatively tight (std dev ~4.26).

Clean separation in the raw data: Looking at individual rows, there's almost no overlap — patients with sugar levels above ~115 and BP above ~120 consistently fall in the "Yes" group, while those below tend to fall in "No." This suggests the four variables together (or even sugar level alone) separate the groups quite well in this sample.

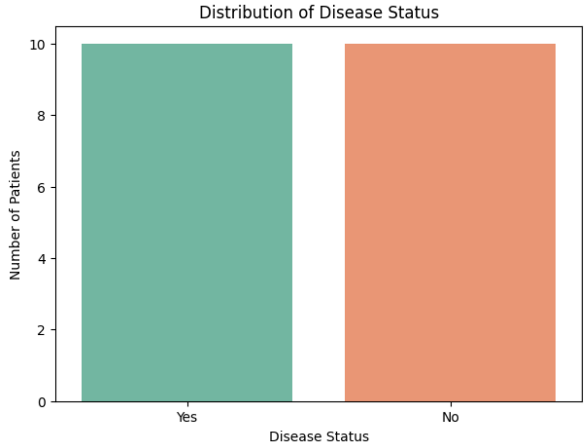
Balanced Interpretation

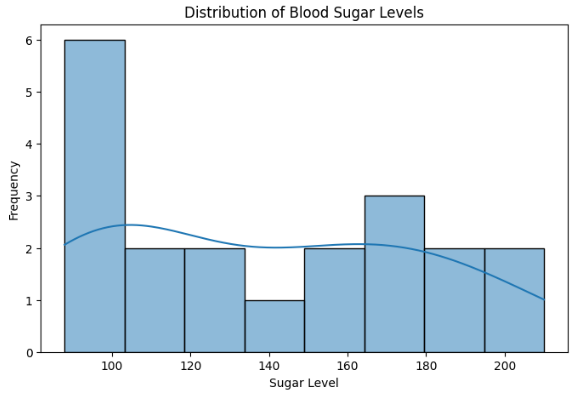
Strengths:

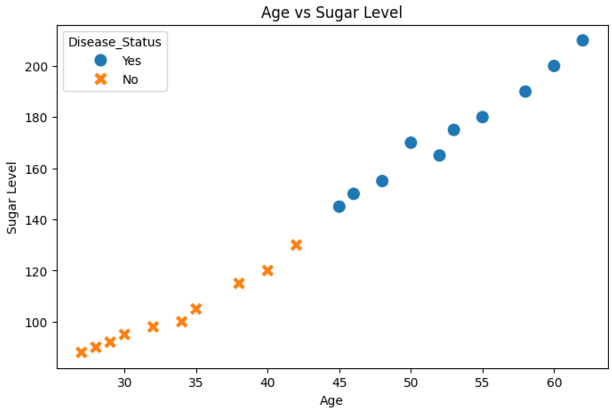
- Clean, complete data with a perfectly balanced outcome variable — a good foundation for analysis or modeling.
- The group-wise averages clearly show directionally consistent, clinically sensible patterns across all four variables.

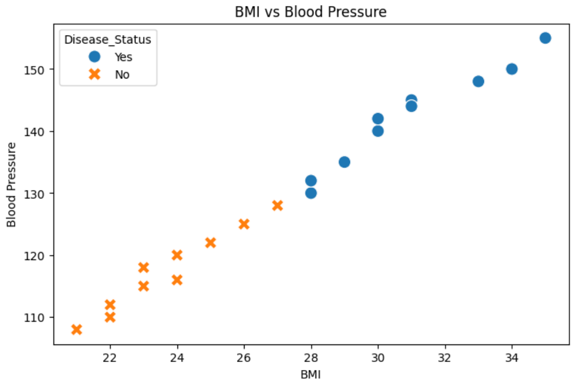
Limitations to flag:

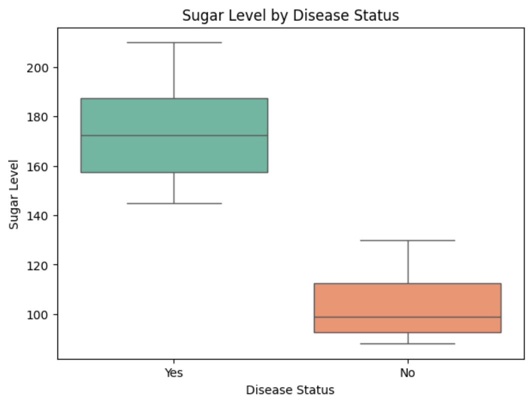
- **Small sample (n=20).** These averages and gaps are suggestive, not statistically robust — a larger cohort could narrow or widen these differences.
- **Correlation, not diagnosis.** This data shows association between these metrics and disease status; it doesn't establish causation, and real diagnosis depends on far more clinical context (family history, other biomarkers, symptoms) than four numbers.
- **No significance testing done yet** — a t-test or ANOVA would confirm whether these group differences are statistically meaningful rather than possibly due to chance in a small sample.
- **Multicollinearity is likely.** Age, sugar, BP, and BMI often move together biologically (e.g., aging affects BP and metabolism), so it's hard to isolate which variable is driving risk versus just correlating with it.
- **This should not be used as a clinical or diagnostic tool** — it's an exploratory statistical summary, not a substitute for medical evaluation.

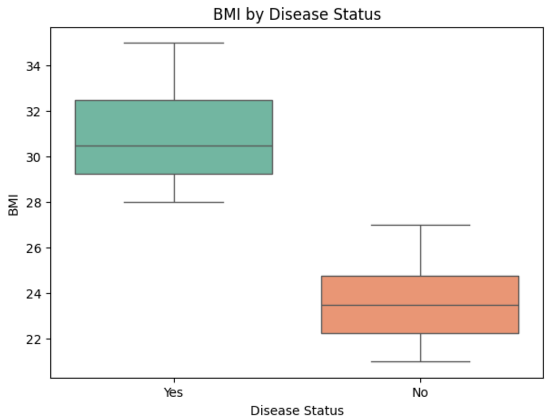


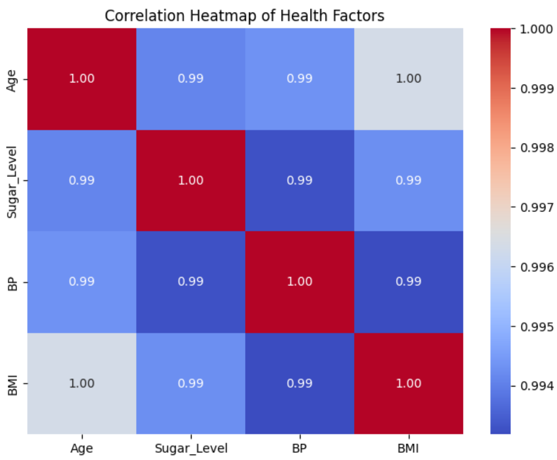












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3 . Exploratory Data Analysis of Retail Sales and Revenue Patterns

Aim

To perform Exploratory Data Analysis (EDA) on retail sales data and identify high-selling products and revenue patterns based on product category, price, quantity sold, and discount using statistical analysis and data visualization.

Dataset Overview

The dataset covers **20 products** across 5 categories (Grocery, Snacks, Dairy, Fruits, Vegetables), with 6 attributes: Price, Quantity Sold, Discount, and Revenue. The data is clean — no missing values, no duplicates.

Key Findings

Overall stats: Average price is ₹54 (range ₹20–100), average quantity sold is 24 units (range 10–45), and average revenue per product is ₹1,114 (range ₹810–1,540). Price and quantity show considerable spread (std devs ~22.80 and ~9.67), meaning products vary widely in both positioning and sales volume.

Category performance diverges sharply between volume and revenue:

Category	Qty Sold	Revenue	Revenue/Unit (approx)
Fruits	85	5,980	~₹70
Vegetables	119	4,400	~₹37
Grocery	57	4,280	~₹75
Snacks	152	4,070	~₹27
Dairy	67	3,550	~₹53

Fruits is the strongest performer — it generates the highest revenue (₹5,980) despite only middling volume, thanks to a high price point. **Snacks sells the most units (152)** by a wide margin, but converts that volume into the lowest revenue of all categories, reflecting its low price per unit. **Grocery is a "low volume, high value" category** — the fewest units sold (57) but still ranks third in revenue, driven by high unit prices. **Dairy is the weakest category** on both volume and revenue.

Top-selling vs top-revenue products barely overlap: The high-volume list is dominated by Snacks (3 of top 5) and one Vegetable product, while the high-revenue list is dominated by Fruits (4 of top 5). This confirms that **volume and revenue are driven by different products** — cheap, high-turnover items (Snacks) versus higher-priced, moderate-turnover items (Fruits).

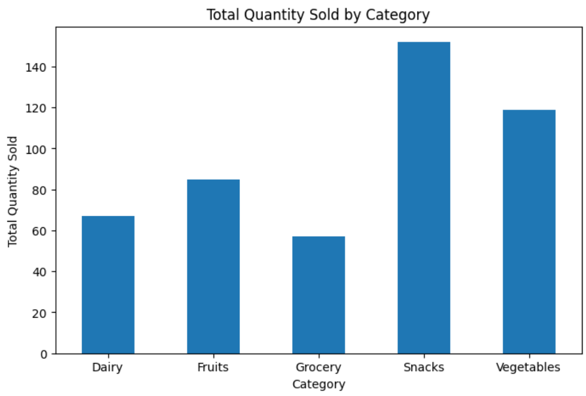
Balanced Interpretation

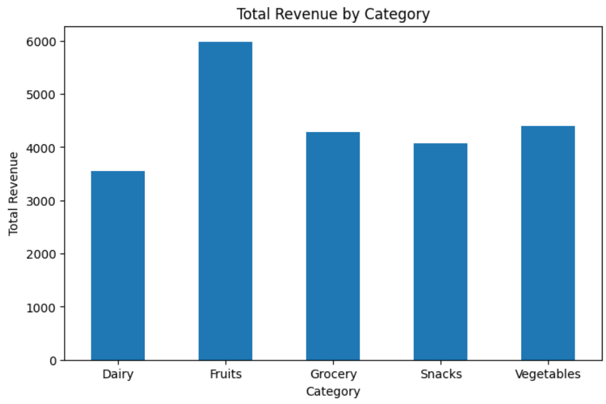
Strengths:

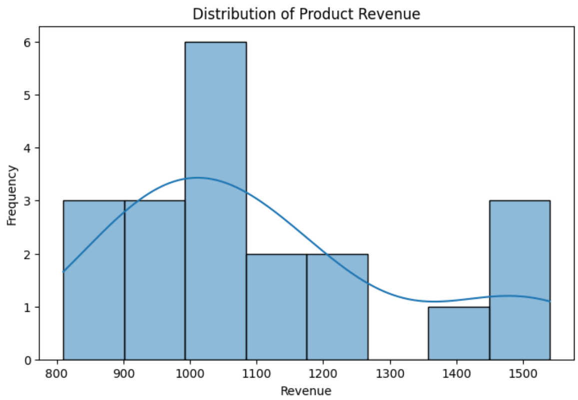
- Clean, complete dataset made straightforward category-level aggregation possible.
- The volume vs. revenue split is a genuinely useful business insight — it shows which categories to push for footfall/turnover versus which to protect for margin.

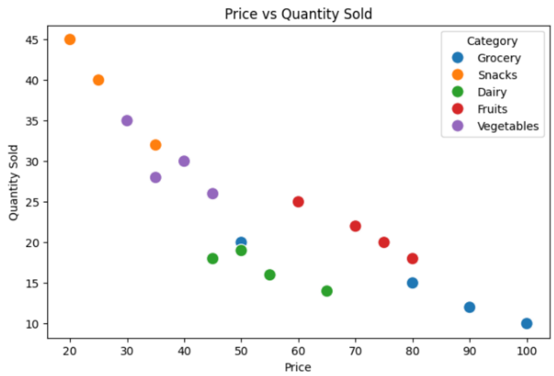
Limitations to flag:

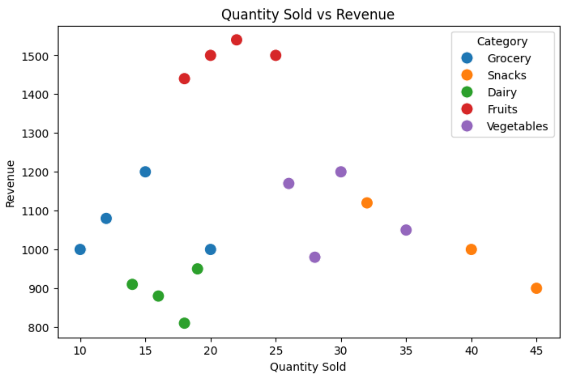
- **Small sample per category (only 4 products each)** — category-level totals could shift substantially with more SKUs, so treat rankings as directional, not definitive.
- **No profit/margin data.** Revenue alone doesn't indicate profitability — a category could have high revenue but thin margins (e.g., if Fruits has higher spoilage/procurement costs), so "best category" conclusions need cost data to be complete.
- **Discount's effect isn't isolated.** Discount ranges 2–7 across products, but this analysis hasn't tested whether higher discounts actually drove more Quantity_Sold — that relationship is unconfirmed.
- **No time dimension.** This looks like a snapshot (or single period) — trends over time (seasonality, growth/decline) aren't visible here.
- **Revenue = Price × Quantity_Sold roughly holds but isn't exact** in a few rows (likely due to discount effects), so revenue conclusions should be read alongside that caveat.

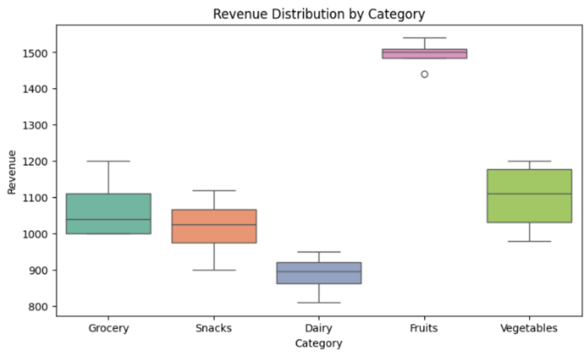












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4. Exploratory Data Analysis of Banking Loan Approval Patterns

Aim

To perform **Exploratory Data Analysis (EDA)** on banking loan data and identify the key patterns associated with **loan approval decisions** based on income, credit score, loan amount, and employment type using statistical analysis and data visualization.

Dataset Overview

The dataset covers **20 customers** across 6 attributes: Income, Credit Score, Loan Amount, Employment Type, and Loan Status. The data is clean — no missing values, no duplicates.

Key Findings

Class balance: A perfectly even split — 10 Approved, 10 Rejected.

Approved applicants outperform rejected ones on every metric, by a wide margin:

Metric	Rejected Avg	Approved Avg	Gap
Income	₹27,200	₹58,800	~₹31,600
Credit Score	584.5	743.5	~159 points
Loan Amount	₹154,500	₹292,000	~₹137,500

Income shows the largest relative gap (more than double), followed by credit score and loan amount. Interestingly, approved customers also requested **larger** loan amounts on average — suggesting approval here tracks applicant strength (income/credit) more than loan size/risk exposure.

Employment Type is a perfect predictor in this sample: All 10 Salaried applicants were approved; all 6 Self-employed and all 4 Unemployed applicants were rejected. There is zero overlap — no Salaried rejections, no non-Salaried approvals.

Correlations among numerical features are extremely high:

- Income–Credit Score: 0.98
- Income–Loan Amount: 0.99
- Credit Score–Loan Amount: 0.95

All three variables move together almost in lockstep.

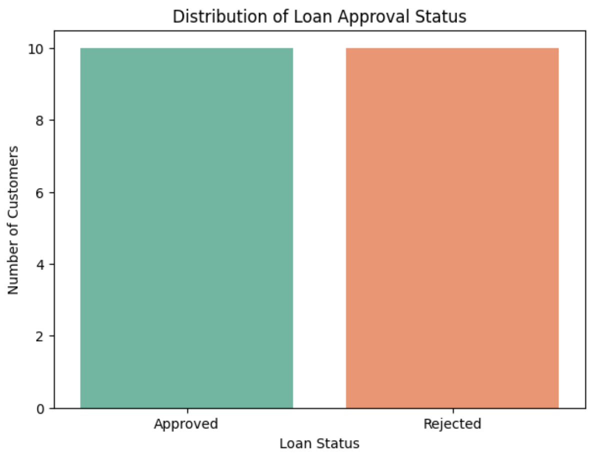
Balanced Interpretation

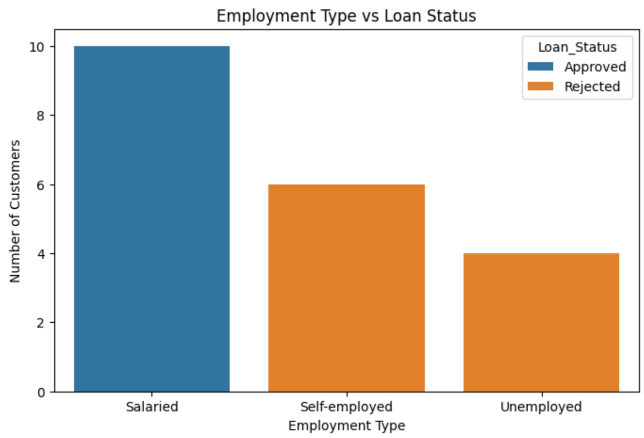
Strengths:

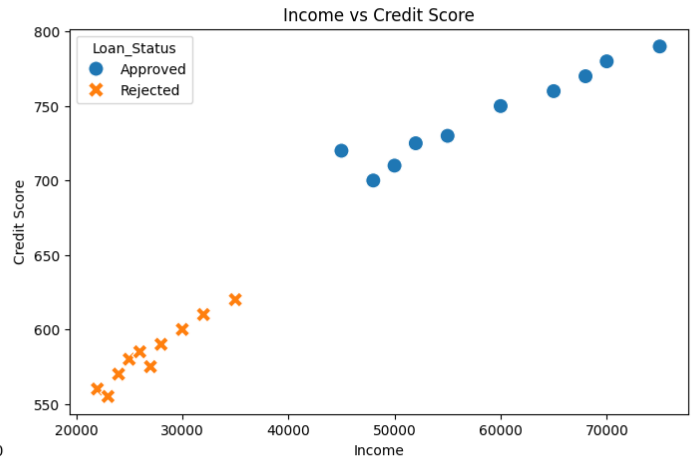
- Clean, complete, balanced dataset — good for illustrating classification concepts.
- The group averages and crosstab clearly show which factors align with approval.

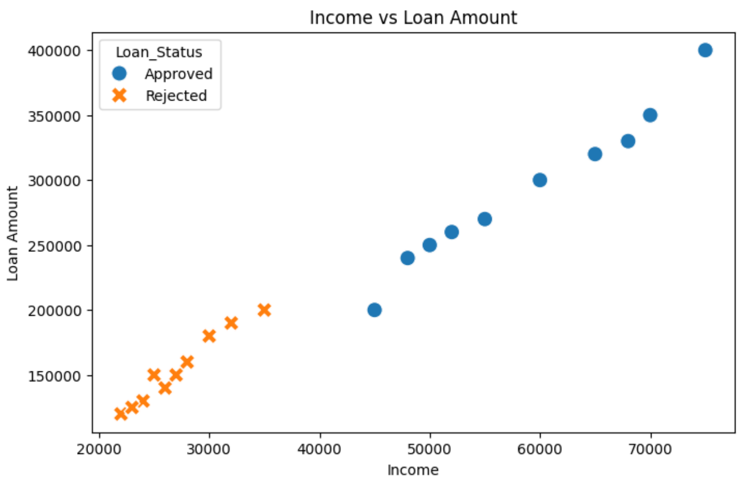
Limitations to flag (these matter more here than in the earlier datasets):

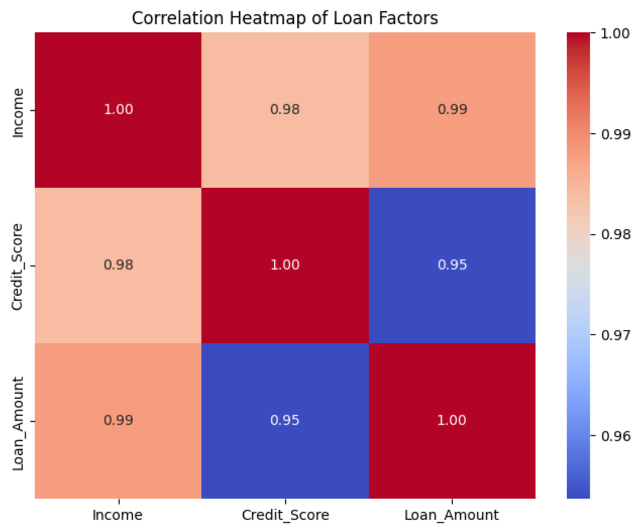
- **The employment-type split is suspiciously perfect.** A 100% clean separation (10/10 and 10/10) on real-world loan data would be unusual — it suggests this may be **synthetic or heavily simplified data**, not a real portfolio. Treat conclusions as illustrative, not representative of actual lending patterns.
- **Correlations of 0.95–0.99 indicate severe multicollinearity.** Income, Credit Score, and Loan Amount are so tightly linked that it's statistically difficult to say which one is "driving" approval — they're essentially moving as a single signal in this sample. Any model built on this data risks unstable or misleading coefficient estimates unless this is addressed (e.g., dropping/combining correlated features).
- **Employment Type may be a proxy, not a cause.** Since it aligns perfectly with income and credit score in this data, it's unclear whether employment type independently matters or simply correlates with the "real" drivers (income/credit stability). Real lending decisions typically weigh many more factors (debt-to-income ratio, repayment history, collateral, existing liabilities) than shown here.
- **Small sample (n=20)** limits how far these patterns can be generalized.
- **Fairness consideration:** if this were a real model, using Employment Type as a near-deterministic approval factor could raise fairness/discrimination concerns (e.g., against self-employed or gig workers) — worth flagging if this dataset informs any actual policy or model design.











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5. Exploratory Data Analysis of Agriculture Crop Yield

Aim

To perform Exploratory Data Analysis (EDA) on agricultural data and identify the key factors influencing crop production and yield based on rainfall, temperature, fertilizer usage, and soil type using statistical analysis and data visualization.

Dataset Overview

The dataset covers **20 farms** across 6 attributes: Rainfall, Temperature, Fertilizer usage, Soil Type, and Crop Yield. The data is clean — no missing values, no duplicates.

Key Findings

Soil type distribution: Fairly even — Loamy (7), Sandy (7), Clay (6).

Soil type shows a dramatic and consistent effect on yield:

Soil Type	Count	Mean Yield	Min	Max
Loamy	7	3,514.29 kg	3,100	3,900
Clay	6	2,941.67 kg	2,700	3,150
Sandy	7	2,021.43 kg	1,800	2,300

Loamy soil is the clear top performer, averaging ~74% higher yield than Sandy soil, with Clay sitting in between. Notably, the ranges don't overlap at all between groups — even the *lowest* Loamy farm (3,100 kg) outproduces the *highest* Clay farm (3,150 kg) is close, and the highest Sandy farm (2,300 kg) is well below the lowest Clay farm (2,700 kg). This is an unusually clean separation.

Rainfall and Crop Yield are almost perfectly correlated (0.99–1.00) — essentially moving together. The Top 5 highest-yield farms are all Loamy and all have the highest rainfall (880–960mm) in the dataset, reinforcing this link.

Temperature is near-perfectly *negatively* correlated with yield (-0.99) — higher temperatures consistently pair with lower yield. Combined with rainfall's positive correlation, this paints a picture where cooler, wetter conditions drive better output.

Fertilizer is also strongly positively correlated (0.98) — more fertilizer aligns with higher yield.

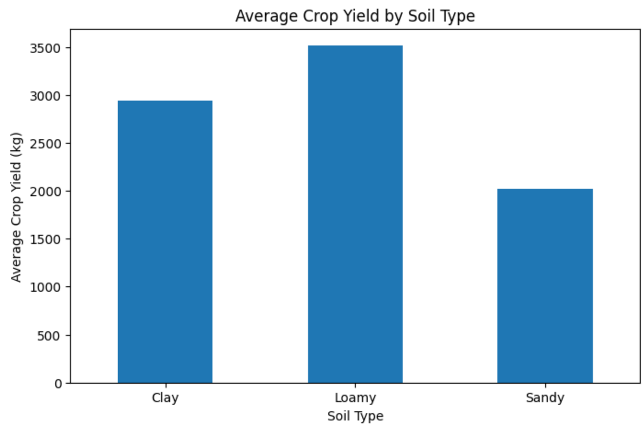
Balanced Interpretation

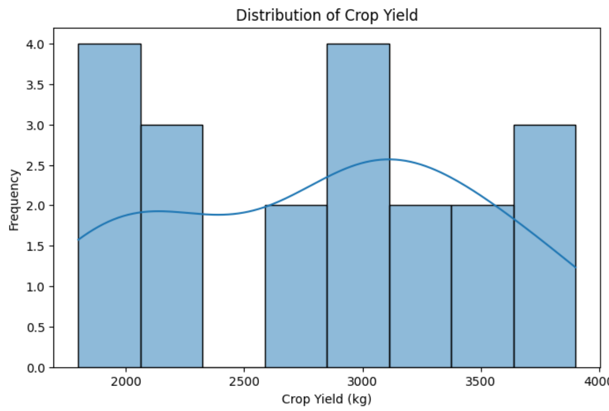
Strengths:

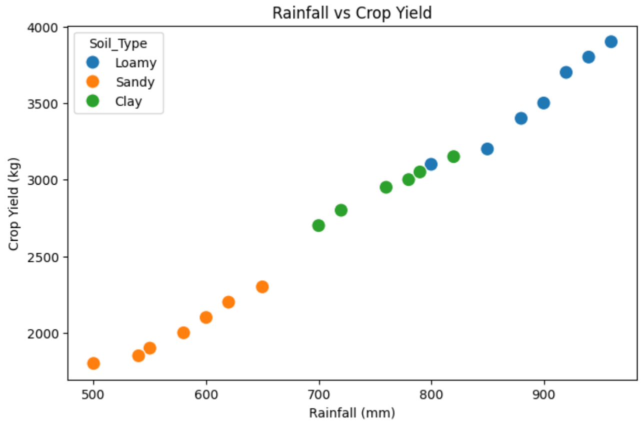
- Clean, complete dataset with intuitive, agriculturally-sensible directional relationships (more rainfall/fertilizer → higher yield; higher temperature → lower yield).
- Soil-type grouping clearly highlights Loamy as the standout performer.

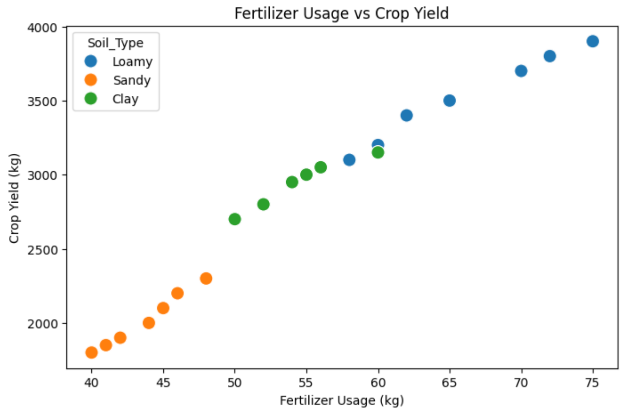
Limitations to flag (important here):

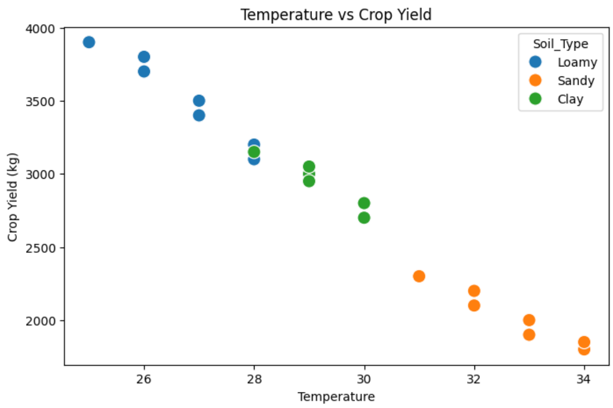
- **Near-perfect correlations (0.98–1.00) across all variables are a red flag for real-world data.** In practice, rainfall, temperature, and fertilizer rarely move in such a tight, near-linear lockstep with yield. This strongly suggests the dataset is **synthetic or simplified for demonstration**, not observed field data. Treat findings as illustrative of relationships, not as validated agronomic conclusions.
- **Severe multicollinearity.** Rainfall, Temperature, and Fertilizer are all highly correlated *with each other* (e.g., Rainfall–Temperature at -0.99, Rainfall–Fertilizer at 0.98), not just with yield. This makes it statistically impossible to isolate which factor is actually "causing" the yield increase — they're confounded with each other and likely also with Soil_Type (Loamy farms tend to have the highest rainfall and fertilizer too, per the Top 5 table).
- **Soil Type may be the real driver, or just another correlated variable.** Since Loamy farms cluster with high rainfall and high fertilizer, it's unclear whether soil type has an independent effect or is simply co-occurring with favorable conditions in this sample.
- **Small sample (n=20, and soil-type subgroups of only 6–7 each)** limits how confidently these patterns generalize.
- **No causal claims should be drawn.** This is correlational, observational data — real yield depends on many additional factors (pest pressure, seed variety, irrigation timing, soil pH/nutrients beyond fertilizer, farming practices) not captured here.

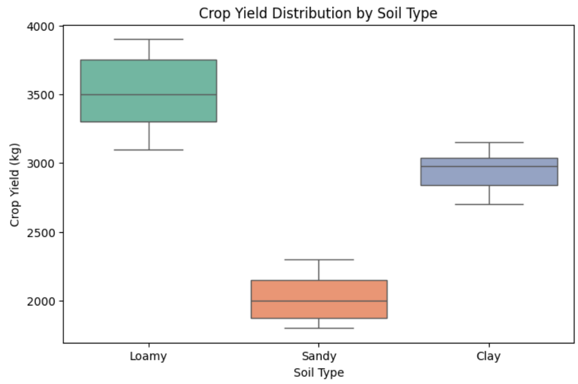


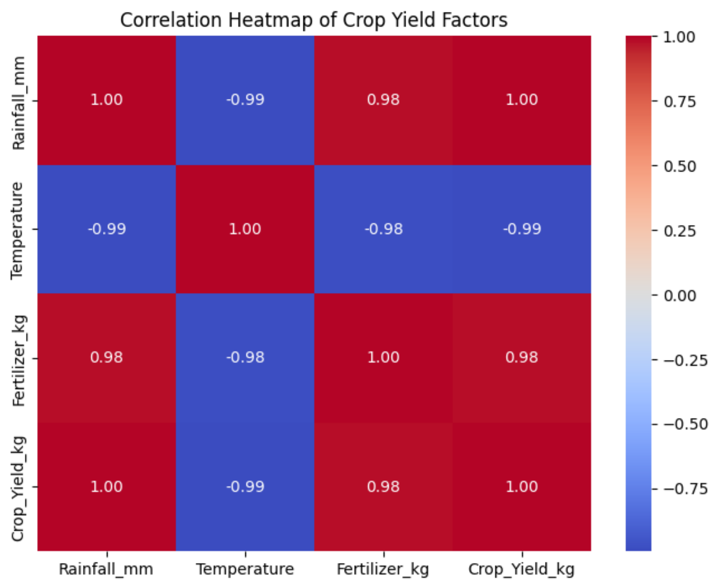












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